

(a) Arrival

electronics package, one of three
(GNSS, resistojets, solar cells)

Kevlar lines to nose and tail
hold the rod's angle; 8 m long,
not to scale

68–75 km/s

spin 23°/s

solid polyethylene rod, 1.10 m by 11 cm,
metal under a clear coat, no electronics

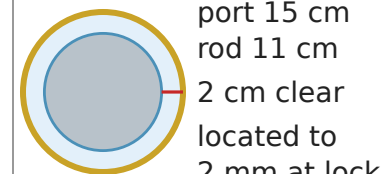
movable insert carries
the port to the intercept

ship's forward face,
standoff foil dashed

hydrogen, 77.7 kg in 80 m³

polyethylene foam plug, 1.56 m deep,
inside a solid polyethylene skin

exhaust throat,
0.69–0.80 m across,
21–28× entrance area



(b) After impact (sketch)

tether lines strike the standoff foil;
packages pass outside the ship

a small share leaks back
out the 15 cm entrance

front spreads as it sweeps the hydrogen

rod and plug turn to plasma together

most gas leaves
through the wide
throat

(c) Port detail for the 2.5 kg rod (enlarged)

2.5 kg rod, 69 cm by 6.9 cm

final trim, up to 5 cm,
in the last 103 ms

electromagnetic lock pulls
the insert onto the backing
20 ms before arrival

linear motors slide the insert in two axes
(third motor out of the page), at most 10 g

900 bar presses the insert onto the backing

foam plug, 0.98 m by 10.9 cm, in a 5 mm
solid skin; it rides on the insert

insert with the 10.9 cm port, shown 6 cm
off center; dashed where centered

fixed backing carries the pulse; its 31 cm
hole spans the insert's stroke